

2 Motor neurones are cells within the nervous system.

(a) Fig. 2.1 shows a diagram of a motor neurone.

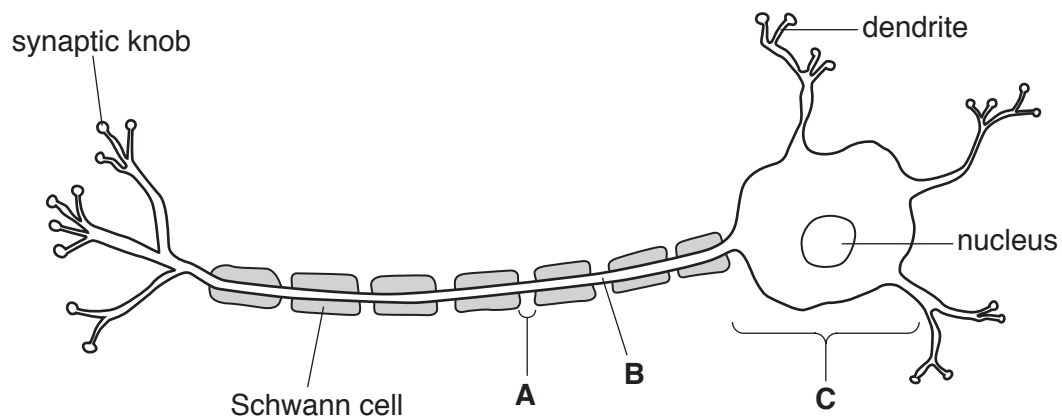


Fig. 2.1

(i) Name the structures labelled **A**, **B** and **C** on Fig. 2.1.

A

B

C [3]

(ii) Describe the function of a motor neurone in a reflex arc.

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..... [2]

- (b) Fig. 2.2 shows a picture of a blue whale, *Balaenoptera musculus*. Blue whales are the largest living mammals and have motor neurones of the type shown in Fig. 2.1. These motor neurones can be up to 30 metres long. The speed of nerve impulses along this type of motor neurone is fast.

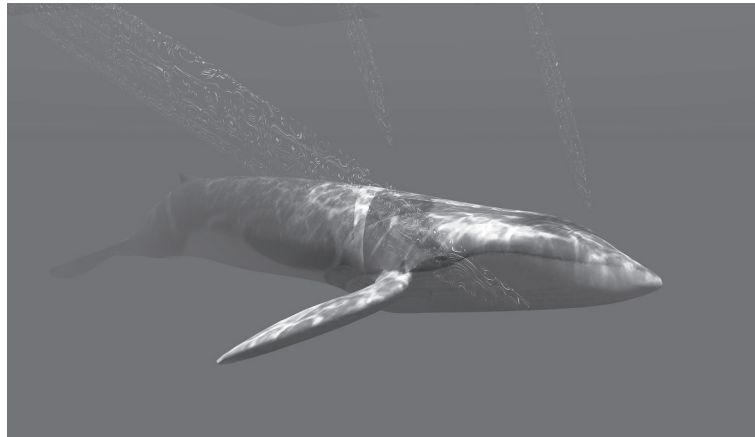


Fig. 2.2

- (i) With reference to Fig. 2.1, explain the fast transmission of impulses along this type of motor neurone.

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- (ii) Suggest why fast transmission of nerve impulses is particularly important in the blue whale.

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[Total: 10]